Classification Task: Decision Tree
Loan Approval Classifier

Machine Learning Assignment 8

October 7, 2026

1 Introduction

In this assignment, we build a Loan Approval Decision Tree Classifier to predict whether an applicant will be approved or denied a loan. Decision Trees are highly interpretable models, allowing us to trace the exact logic and thresholds used by the model to make lending decisions.

2 Decision Tree Visualization

We constrained the tree to a maximum depth of 3 to maintain readability. The visualization below shows how the algorithm splits the data based on feature thresholds, separating the classes into "Approved" and "Denied" leaf nodes.

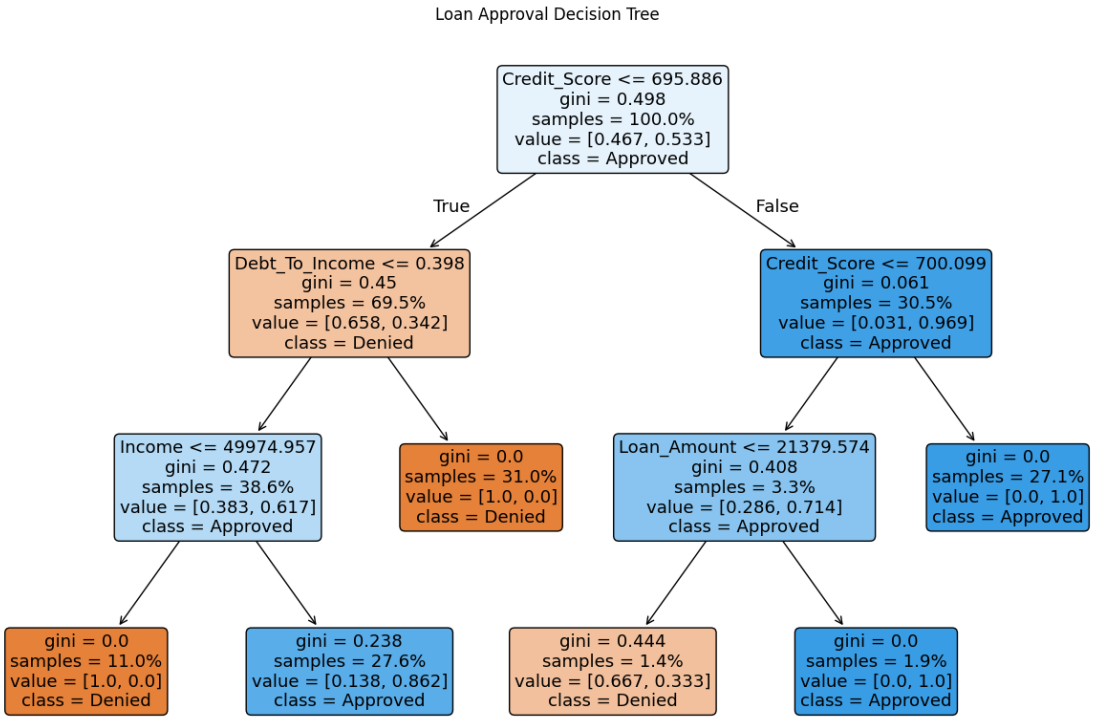


Figure 1: Visualized Loan Approval Decision Tree.

3 Top 3 Features and Interpretation

Based on the internal feature importances (Gini importance) calculated by the model, the top 3 features relied upon for the loan approval decisions are:

1. **Credit Score (Importance: ∼40.2%):** This is the most critical feature. The tree uses it heavily at the root and subsequent splits. In real-world lending, a high credit score is the strongest indicator of a borrower's historical reliability and ability to repay debt, making it the primary filtering criterion for approval.

2. **Debt-To-Income Ratio (Importance: ∼30.7%):** The second most important feature. Even if a borrower has a decent credit score, a high debt-to-income (DTI) ratio indicates they are currently overburdened with debt relative to their earnings. The tree correctly learns to deny loans to individuals whose DTI exceeds safe thresholds.

3. **Income (Importance: ∼27.4%):** The third top feature is absolute income. While DTI is crucial, absolute income provides the baseline for how much financial buffer a borrower has. The model uses income to make finer distinctions, especially for borderline applicants.

These top 3 features exactly mirror real-world lending intuition: lenders primarily care about your financial history (Credit Score), your current capacity to take on new debt (Debt-To-Income), and your raw earning power (Income). The features *Years Employed* and *Loan Amount* were deemed much less significant by the model in this specific dataset context.